

Juan David Muñoz Bolaños

Rechenauerstraße 42, Innsbruck, Österreich

+43 676 9115145

jmunozbolanos@gmail.com

April 21, 2026

Carl Zeiss Vision International GmbH

Turnstraße 27

73430 Aalen

Germany

Pionier für Extended Reality (XR) | Ref. JR_1048257-1

I was excited to discover the Pioneer Programme for Extended Reality at ZEISS/tooz. As a Physics Engineer with a deep passion for optical innovation, I see the development of smart glasses as the perfect convergence of high-precision optics and wearable technology — a frontier I am eager to help define.

Throughout my career I have specialized in bridging complex optical prototypes to robust, production-ready systems. During my PhD at the Medical University of Innsbruck I built and integrated a full adaptive optics platform spanning Texas Instruments DMD light modulators, high-speed cameras, laser sources, and FPGA-based real-time control. I also developed SpectraMap, an open-source Python package for automated spectral data processing, reflecting my drive to build scalable, reusable tools alongside hardware work.

My background in wavefront metrology and adaptive optics offers a direct bridge to the vision-correction and display challenges unique to AR/XR systems. I am comfortable driving projects end-to-end — from feasibility and component selection through commissioning — and I thrive in the kind of interdisciplinary, fast-moving environment that pioneering a new product category demands.

The opportunity to shape the future of mobile computing platforms at a world-leading optical company like ZEISS is a compelling one. I am available from June 2026 and would welcome the chance to discuss how my experience can contribute to the tooz Extended Reality vision.

Kind regards,

Juan David Muñoz Bolaños

PhD Candidate, Medical University of Innsbruck

Juan David Muñoz Bolaños

Machine Vision Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

jmunozbolanos@gmail.com | LinkedIn Profile



Summary

Physicist and Machine Vision Engineer specializing in industrial automation and precision optical system development. Expertise bridging robust 2D/3D inspection prototypes to seamless plant software integration via C++, Python, and Halcon.

Professional Experience

Jan'22 | Present **Med. Univ. Innsbruck**

Innsbruck, Austria

Optical Systems Engineer | PhD Researcher

Closed-Loop Optical Aberration Correction (ACS Photonics): Implemented a Python-driven adaptive optics feedback loop achieving diffraction-limited confocal imaging in scattering biological tissue using a laser, camera, and CCD sensor, and light modulator.

Multi-Device Adaptive Optics Microscope (Biomedical Optics Express):

Designed and integrated a two-photon imaging platform spanning FPGA, NI LabVIEW, high-speed galvos, Basler cameras, and Arduino control.

Interferometric Wavefront Metrology Platform: Automated phase retrieval with a high-speed holographic modulator of Texas Instruments via a C++, Python, and LabVIEW-FPGA stack to restore diffraction-limited imaging quality.

Mar'18 | Mar'19 **INTECOL S.A.S.**

Medellín, Colombia

Junior Machine Vision System Engineer

SICK Sensor Integration for Automated Filling Control: Commissioned SICK photoelectric and distance sensors on a filling line; integrated output signals into the PLC loop for repeatable dispense.

Cognex Smart Camera for Automotive Robot Guidance: Deployed a Cognex vision system to identify car bumper variants and feed positional data to a robotic arm for selective chemical surface treatment.

Halcon OCR Inspection on High-Speed Bottle Lines: Built a Halcon-based vision system with flash illumination and adaptive segmentation to verify label codes at full line speed despite surface variation.

Technical Competencies

Industrial Vision Software: Halcon (MVTec), Cognex VisionPro, OpenCV (C++/Python). Experience in pattern matching, OCR, blob analysis, and adaptive segmentation for high-speed inspection lines.

Cameras, Sensors & Illumination: Basler and IDS CMOS/CCD cameras, telecentric lenses, SICK sensors; flash, bright-field, dark-field, and ring-light illumination design.

Programming & Automation: C++, Python, LabVIEW/FPGA for system control, image processing pipelines, and PLC signal integration. Experience with real-time data acquisition and instrument synchronization.

Optical Design & Metrology: Knowledge ZEMAX OpticStudio, Python modelling and tolerance analysis. Interferometric wavefront sensing, adaptive

optics, and precision opto-mechanical system integration (Thorlabs, Edmund Optics, Newport).

Feasibility & Lifecycle: End-to-end: requirements, feasibility studies, procurement, production-floor commissioning, and Gage R&R validation.

Education

Jan'22 Jun'26	Medical University of Innsbruck <i>Doctor of Philosophy in Adaptive Optics & Precision Metrology</i>	<i>Innsbruck, Austria</i>
2019 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	<i>Jena, Germany</i>
2012 2018	University of Cauca <i>Bachelor of Science in Physics Engineering</i>	<i>Popayán, Colombia</i>

Publications & Awards

Awards: Selected for ZEISS SMT, and SPIE Photonics West(San Francisco) Conference. COLFUTURO 2016-2018 (best Colombian professionals).

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills & Hobbies

Ownership & Initiative: Lead project focus from feasibility to production floor commissioning (e.g. SpectraMap).**Teamwork & Analysis:** Proven collaboration across optics, hardware, and software disciplines with structured root-cause analysis of defects.

Hobbies: Enjoying the mountains (Biking, Hiking, learning Skiing) • Bachata Dancing • Harmonica • Vision Systems

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte <i>Medical University of Innsbruck</i> monika.ritsch-marte@i-med.ac.at	Prof. Dr. Alexander Jesacher <i>Medical University of Innsbruck</i> alexander.jesacher@i-med.ac.at
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INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
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PBX. (574) 316 3322 Cell: (57) 301 430 2532

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

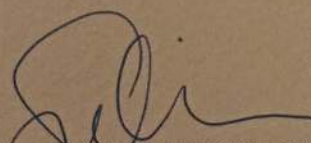
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

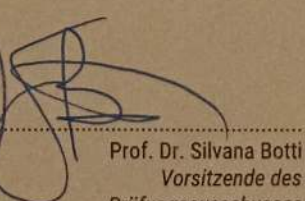
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum
BFI Tirol Bildungs GmbH
in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No.

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry
10 DIC/DEC 2028

Autocidad / Authority

G. ANTIOQUIA

Forma dei bulbi / *bulbos* e *seguenti*

Quar. Family

[illegible]

Stadtmagistrat Innsbruck
MA II - Bezirks- und Gemeindeverwaltung
Standesamt und Personenstandsangelegenheiten
Aufenthaltsangelegenheiten
post.aufenthaltsangelegenheiten@innsbruck.gv.at

Zahl: Maglbk/58566/AUFG-NAG/3/1 – 00053337-04

Antragsprotokoll

Herr: **Muñoz Bolaños Juan David**
wohnhaft in: **6020 Pradl, Reichenauer Straße 42/Top 5**
geboren am: **04.06.1994**
Reisedokument-Nr.: **AV410884**
Nationalität: **Kolumbien**
hat am: **13.03.2026**

beim Stadtmagistrat Innsbruck, Referat Aufenthaltsangelegenheiten, einen Zweckänderungsantrag auf Rot-Weiß-Rot - Karte plus eingebracht.

Ein rechtzeitig eingebrachter Verlängerungsauftrag räumt während der gesamten Dauer des Verlängerungsverfahrens dieselbe Rechtsposition ein, die nach Erhalt des letzten Aufenthaltstitels innegehabt wurde. Dies bedeutet auch, dass bei befristet erteilten Aufenthaltstiteln mit unbeschränkten Zugang zum Arbeitsmarkt („Rot-Weiß-Rot Karte plus“ und „Familienangehöriger“) weiterhin während der gesamten Dauer des Verlängerungsverfahrens ein Zugang zum Arbeitsmarkt besteht. Bei darüber hinaus gehenden Fragen einer allfälligen weiterhin bestehenden Beschäftigung wenden Sie sich bitte an das Arbeitsmarktservice (AMS).

Nachzureichende Unterlagen: ***

Die Erledigung des vollständig eingebrachten Antrages wird voraussichtlich innerhalb von Wochen erfolgen.

Persönliche Abholung (erforderlich !) jedenfalls erst ab: 17.05.2026, Zimmer: .Nummer ziehen

Zur Abholung (Abholzeiten: Montag bis Freitag, 08.00 bis 11.30 Uhr) sind mitzubringen:

- **der Reisepass**
- **Aufenthaltstitel in Kartenform**

Für den Bürgermeister:

(Huter)

ausstellende Behörde:

Stadtmagistrat Innsbruck
Abteilung II
Aufenthaltsangelegenheiten

übernommen am: 13.03.2026

Unterschrift:

Bei überprüfenden Amtshandlungen drei Monate nach Ausstellung dieser Bestätigung wird um Rückfrage bei der zuständigen Aufenthaltsbehörde ersucht. (Tel.: 0512/5360-1030/1032/1034/1036)